IMDB 3.0

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Abstract

Streaming recommendation systems are highly effective at predicting patterns in user behavior, yet they still struggle with one fundamental limitation: they cannot know why a user likes a particular movie without meaningful input. Even with advanced matrix factorization and deep neural network pipelines. These systems repeatedly surface similar content because they infer intention from limited signals such as clicks, watch time, or genre tags. This is a gap, not a failure of the algorithms, but a lack of user controlled context leading to constant cases of endless scrolling. Intuitively speaking when looking for a new movie you need to have the option to ask multiple questions to hone in on what you like the most.

Our project aims to solve this problem by giving users direct influence over the recommendation process; we extract visual, textual, and audio embeddings from official movie trailers using CLIP, Gemini, and Librosa, producing three interpretable vectors for each film. Through a late fusion approach users can specify which aspect of a movie matters to them, allowing the system to generate recommendations based on their chosen criteria rather than hidden model assumptions. With a custom trailer data set and integrated OMDB meta data, the result is an interactive platform that offers transparent vectors, controllable similarity search, and a practical and scalable alternative to passive recommendation feeds.

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1 Introduction (1 page)

People often struggle to find a movie to watch. it is very easy for a streaming service user to start doom scrolling on their for you page and never end up finding anything they want to watch. We believe that this is because most existing systems rely on limited behavioral signals, which restricts how accurately they can predict user intent. Most existing systems rely on a variety of features; User Taste, Time of day, and many more. Even though many well thought out systems have been put into place, people often start getting recommended the same things over and over again or have a hard time finding new things. The reason why is because at the end of the day there is no definitive success signal to prove that their algorithm is working. Moreover, the feeling of being recommended the same thing over and over again comes from the nature of the Matrix Factorization.

Many users get frustrated with this and decide to move to different platforms for this reason alone. So to tackle this issue we sought to emulate the feeling of asking someone for a movie recommendation.

We believe that asking someone who is knowledgeable about many movies in the past is the best way to find new movies and learn more about what you really like

Here is a scenario of that; if you walked into a movie store and ask the store owner for a movie recommendation, they typically would follow up with "Well, What type of movie do you like" from this you can start to express your favorite movie and why you like it "I like Wal lee, I don't really care for the realistic graphics but I really like the story told and the Scifi setting". Then from there the store owner may ask more questions to hone in on what you like more, from that they may start recommending movies to you then and there.

from this interaction the user gains more than just movie recommendations that they may like they also learn what type of movies they like and why.

That feeling of truly understanding why you like a certain genre and being able to express to someone knowledgeable in movies if they can find movies like that is what we are trying emulate by using vector similarity.

Using CLIP, Gemini and Librosa we are able to vectorize movies into 3 distinct categories from their trailers. Audio, Visuals and Narrative. from this we created vectors out of the text descriptions from these categories. Now to let the user start of their search they have to click on a movie they like, from their they can mess with the vectors unique to that movie through sliders to find movies that capture those same vibes semantically.

There are many vector similarity search engines that currently exist, but they do not allow the user to control their movies vector like our system does which eliminates the control and exploration in discovering new movies.

We hope that our Movie Recommendation system will allow people to learn more about what type of movies they truly like, and to start developing their own unique taste in movies. Hopefully we are able to shine a light on other under appreciated movies that may go under the radar due to current recommendation systems.

Our work contributes to a new approach to recommendation systems combining multi modal trailer embeddings with direct user controlled vector manipulation. Instead of treating preference prediction as a closed problem, we open the process to the user, allowing them to guide the system towards what they actually care about. This bridges the gap between passive algorithmic inference and an active interpretable response experience. While existing systems already preform large scale prediction effectively, our goal is not to rebuild or replace them, but to extend them with the missing layer of explicit user control, adding content that current models cannot infer on their own.

1.1 Research Question(s)

Our project addresses three primary research questions that challenge the current paradigm of passive recommendation systems:

RQ1: Can we give users explicit control over recommendation criteria?

Traditional recommendation systems rely on implicit behavioral signals (clicks, watch time, browsing patterns) to infer user preferences. However, these signals are inherently ambiguous a user might watch a movie for its visual spectacle, compelling narrative, or atmospheric soundtrack, but the system cannot distinguish which aspect drove their interest. Our first research question asks: *Can we design a system where users explicitly manipulate the features that drive recommendations, rather than having the system guess their intent?*

This question is significant because it addresses the fundamental "black box" problem in recommender systems. Even when algorithms perform well statistically, users often feel frustrated when they cannot understand or influence *why* they received certain recommendations. By decomposing movies into interpretable dimensions (visual, audio, narrative) and allowing users to adjust their relative importance, we hypothesize that users will gain both better recommendations and a deeper understanding of their own preferences.

RQ2: Do multimodal trailer embeddings provide interpretable dimensions for similarity?

Our second research question investigates whether modern deep learning models can extract meaningful and separable representations from movie trailers. Specifically: *Can visual (CLIP), audio (Librosa), and narrative (Gemini + BERT) embeddings capture distinct aspects of movies that align with human intuition about similarity?*

This is important because not all embedding approaches produce interpretable dimensions. If our three modalities simply encode redundant information (e.g., visual and narrative both primarily capture genre), then giving users control over their weights would be meaningless. Conversely, if each modality captures a genuinely different aspect cinematography vs. plot structure vs. sonic atmosphere then users can leverage this separation to express nuanced preferences like "I want movies with similar visual style but different narratives."

RQ3: Does user control improve the movie discovery experience?

Finally, we ask the practical question: *Does giving users control over embedding weights lead to more satisfying movie discovery compared to traditional static similarity search or passive recommendation feeds?*

This question evaluates whether our approach addresses the real world problem of "doom scrolling" on streaming platforms. We hypothesize that users who can actively explore the recommendation space adjusting sliders to emphasize different aspects of a seed movie will experience:

- Reduced decision paralysis from endless scrolling
- Discovery of movies they wouldn't find through behavioral algorithms alone
- Greater sense of agency and transparency in the recommendation process
- Better understanding of what they actually like in movies (e.g., "I realize I prefer sci fi settings more than realistic visuals")

1.2 Target Audience

Our target audience are movie watchers who want to have a more intuitive recommendation system, These are users who want more control and transparency over how recommendations are made, rather than passively receiving suggestions soley on historical behavior.

The secondary audience could include film enthusiasts and critics who want to explore films beyond mainstream popularity, as well as researchers who are interested in user centered recommendations systems that use multi modal embeddings techniques.

2 Related Work

Prior work has demonstrated that ... [Zheng et al., 2024].

Zheng et al. [2024] found...

3 Methods

3.1 Dataset(s) & Preprocessing

3.1.1 Dataset Creation

We created a custom movie trailer dataset since no existing dataset met our requirements for official, high quality trailers with consistent metadata. Our data collection pipeline consisted of:

Collection Tools:

- **Selenium**: Automated browser interaction for navigating YouTube
- **YouTube Official API**: Retrieved official trailer videos and metadata
- **yt dlp**: Downloaded video files from YouTube
- **Fuzzy string matching**: Matched movie titles between YouTube and OMDB databases

Data Sources:

1. **YouTube Official Channels**: Paramount Pictures and 20th Century Studios official channels
2. **OMDB API**: Movie metadata including titles, ratings, genres, descriptions, release years, and poster thumbnails

3.1.2 Dataset Composition

Our final dataset contains 250 official movie trailers. Each entry includes:

- Movie title
- YouTube trailer URL
- Downloaded video file (.mp4)
- OMDB metadata (rating, genre, description, year, poster URL)

Table 1: Data Dictionary

Variable	Type	Description
movie_title	String	Official movie title
trailer_url	String	YouTube video URL
video_file	Path	Local path to downloaded trailer
rating	Float	IMDB rating (0 10)
genres	List[String]	Movie genres
description	String	Plot summary
year	Integer	Release year
poster_url	String	Movie poster image URL
visual_embedding	Array[512]	CLIP visual embedding
audio_embedding	Array[384]	Audio feature embedding
narrative_embedding	Array[384]	Narrative text embedding

3.1.3 Preprocessing Steps

1. Video Preprocessing:

- Extracted frames at 1 fps to capture visual diversity
- Resized frames to 224×224 pixels for CLIP input
- Separated audio track from video for audio analysis

2. Audio Preprocessing:

- Extracted audio as WAV format at 22,050 Hz sample rate
- Computed audio features using Librosa:
  - Mel frequency cepstral coefficients (MFCCs)
  - Spectral contrast
  - Chroma features
  - Zero crossing rate
- Aggregated frame level features using mean and standard deviation

3. Text Preprocessing:

- Used Gemini 2.5 Flash to generate narrative descriptions from trailers
- Prompted model to describe: plot themes, storytelling style, tone, pacing, and emotional arc
- Cleaned generated text (removed special characters, normalized whitespace)
- Tokenized for BERT embedding

4. Quality Control:

- Removed trailers shorter than 30 seconds or longer than 5 minutes
- Filtered out non English trailers
- Verified OMDB metadata match accuracy using fuzzy matching threshold of 0.85
- Manually reviewed mismatches

5. Embedding Generation:

- **Visual**: CLIP ViT B/32 model generated 512 dimensional embeddings per frame, averaged across trailer
- **Audio**: Developed our own Digital Signal Processing (DSP) pipeline using the Librosa library encoding into 384 dimensional embeddings
- **Narrative**: DistilBERT encoded generated descriptions into 384 dimensional embeddings
- Normalized all embeddings to unit length for cosine similarity computation

3.2 EDA & Feature Engineering

3.2.1 Dataset Statistics

While our primary data consists of video files rather than traditional tabular features, we conducted exploratory analysis on the metadata and extracted embeddings to ensure dataset quality and understand its characteristics.

3.2.2 Data Quality Assessment

Download Success Rate:

- Initial dataset: 250 movies identified from official studio channels
- Download success rate: 100% (some videos were restricted/private/unavailable)
- Final processed dataset: 250 trailers with complete multimodal embeddings
- Failed downloads primarily due to: 403 Forbidden errors, private videos, or unavailable content

Data Collection Robustness:

- Implemented retry logic with exponential backoff (up to 3 attempts per video)
- Used randomized user agents and delays to avoid rate limiting
- Applied fallback strategies for restricted content (Android client, lower quality formats)
- All successfully downloaded trailers were official studio releases from verified channels

Completeness:

- 90% of downloaded trailers had complete OMDB metadata
- Missing metadata handled by excluding entries from final dataset or manual edits
- All retained entries have: title, genre, rating, description, poster URL, and all three embeddings

Video Quality:

- All trailers verified as official studio releases
- Preferred resolution: 720p 1080p (format codes 95/94/93/92/91)
- Standardized output format: MP4 with H.264 video codec

Audio Quality:

- Sample rate standardized to 22,050 Hz
- Trailers with corrupted audio tracks excluded from final dataset

3.3 Models

Our architecture does not rely on a single end-to-end model but rather a composite pipeline of three distinct neural pathways and signal processing algorithms. This "Late Fusion" architecture allows us to process different modalities independently before aggregating them during retrieval.

3.3.1 Model 1: Visual Encoding (CLIP ViT-B/32)

For visual analysis, we utilized **OpenAI's CLIP (Contrastive Language-Image Pre-training)**, specifically the `ViT-B/32` variant. CLIP is a multi-modal model trained on 400 million image-text pairs, allowing it to understand visual concepts (e.g., "noir lighting," "action sequence") without explicit labeling.

Architecture & Implementation:

- **Input Processing:** We sampled frames from the video trailers at a rate of 1 frame per second (1 fps). Each frame was resized to 224×224 pixels.
- **Feature Extraction:** Each frame is passed through the Vision Transformer (ViT) backbone. We extract the output from the penultimate layer to obtain a 512-dimensional embedding vector per frame.
- **Temporal Aggregation:** To create a single representation for the entire video, we apply mean-pooling across all frame vectors.
- **Color Histogram Fusion:** To capture stylistic color choices, we computed HSV color histograms (96 bins) for each frame.

3.3.2 Model 2: Narrative Understanding (Gemini 2.5 Flash & S-BERT)

Narrative extraction is a two-stage process combining a Large Multimodal Model (LMM) for reasoning and a Sentence Transformer for embedding.

Stage 1: Cognitive Processing (Gemini 2.5 Flash) We employed Google's **Gemini 2.5 Flash** model via the Generative AI API. Unlike traditional metadata scrapers, we uploaded the full MP4 video file to the model's context window. Using a specific "Director's Cut" system prompt, we instructed the model to analyze the trailer and output structured tags, specifically describing the '[VISUAL_STYLE]', '[NARRATIVE_ARC]', and '[EMOTIONAL_VIBE]'.

Stage 2: Semantic Embedding (all-MiniLM-L6-v2) The structured text generated by Gemini was then fed into **Sentence-BERT (S-BERT)**, specifically the `all-MiniLM-L6-v2` model. This is a distilled BERT model optimized for semantic search. It converts the variable-length text description into a dense, fixed-size **384-dimensional vector**.

3.3.3 Model 3: Audio Signal Processing (DSP & Librosa)

For audio, we found that standard pre-trained models often failed to distinguish between movies with similar loudness profiles. Therefore, we developed our own Digital Signal Processing (DSP) pipeline using the **Librosa** library.

Feature Engineering: Instead of a simple global average, we implemented **Temporal Slicing**, splitting the audio track into 8 distinct time chunks to capture the pacing arc of the trailer (e.g., quiet intro building to a loud climax). For each chunk and the global track, we extracted:

- **MFCCs (Mel-Frequency Cepstral Coefficients):** To capture timbre and differentiate speech from music.
- **Spectral Contrast:** To measure the "texture" of the sound (distinguishing clean dialogue from noisy explosions).
- **HPSS (Harmonic-Percussive Source Separation):** To mathematically separate musical scores from percussion/sound effects.
- **Chroma Features:** To capture harmonic content and pitch class profiles.

- **Zero-Crossing Rate & Entropy:** To measure signal complexity and noisiness.

These features were concatenated and normalized into a high-dimensional **384-dimensional audio fingerprint**.

3.4 Training and Evaluation Methodology

Since our system relies on unsupervised retrieval rather than supervised classification, traditional "training" (gradient descent) was not performed on the embedding models themselves. We utilized a Transfer Learning approach, leveraging pre-trained weights for CLIP and BERT.

Evaluation Method: Evaluating a "vibe-based" recommendation system is inherently qualitative, as there is no ground truth for "visual similarity." We employed the following evaluation techniques:

1. **Cosine Similarity Analysis:** We mathematically verified differentiation by calculating the cosine similarity between trailers of disparate genres (e.g., Action vs. Animation). A successful model should yield lower similarity scores (< 0.85) for distinct genres.

2. **t-SNE Visualization:** We used t-Distributed Stochastic Neighbor Embedding (t-SNE) to project our high-dimensional vectors into 2D space. We evaluated performance by observing if movies of similar genres (e.g., Horror) naturally formed clusters without explicit supervision.

3. **Sanity Checks:** We performed manual queries (e.g., "Find movies that look like *Ron's Gone Wrong*") and qualitatively assessed if the retrieved results shared the query's aesthetic or narrative properties.

4 Results

4.1 Model Evaluation

Our analysis of the extracted embeddings demonstrated that the multi-modal approach successfully disentangled the different aspects of movie content.

Differentiation Capabilities: Initial iterations of our audio model (using simple averaging) resulted in similarity scores of ≈ 0.99 across the dataset, making retrieval useless. After implementing our advanced temporal slicing and HPSS logic, we achieved significant differentiation.

- **Action vs. Animation:** Comparing *The King's Man* (Action) vs. *Bob's Burgers* (Animation) yielded a similarity score of roughly **0.92**. While still high due to the nature of trailer compression, this variance allows the HNSW index to effectively rank distinct soundscapes.

- **Visual Clustering:** The t-SNE projections of the CLIP embeddings showed distinct clusters where "dark/thriller" movies (e.g., *A Haunting in Venice*) grouped separately from "bright/comedy" movies (e.g., *Vacation Friends 2*), confirming that the visual vectors captured lighting and tone.

4.2 Comparative Analysis

We compared our **Late Fusion** approach against a traditional **Early Fusion** baseline (where visual, audio, and text features are concatenated into a single vector before indexing).

Early Fusion (Baseline): In our tests, Early Fusion often led to "signal dilution." For example, searching for a movie with a "Dark Visual Style" but a "Comedic Plot" became impossible, as the two conflicting signals averaged out into a generic vector that returned mediocre results for both.

Late Fusion (Our Approach): Our system creates three independent vector spaces. By querying `db_visuals`, `db_audio`, and `db_narrative` separately and merging the scores with user-defined weights (via UI sliders), we achieved:

- **High Interpretability:** We can explain exactly *why* a movie was recommended (e.g., "95% Visual Match").
- **Modal Independence:** High similarity in one modality (e.g., Audio) does not force a match if the user prioritizes another (e.g., Narrative).

This confirmed that maintaining independent vector indices is superior for exploratory media discovery where user intent varies dynamically.

5 Discussion

5.1 Connections to Prior Work

Our work expands on or connects to previous research that we looked into.

- **The "Why"**: Previous work has been able to relate create a video-based recommendation system for movie trailers (Hsieh et al., 2018), but the reasoning behind why two movies might be similar is not interpretable, and is simply a black box to the user. Our project allows a user to not only understand why movies are related and are being recommended to them, but also grants the capability to customize the weights of certain tags, allowing the user to have a say over how a movie should be considered "related".
- **YouTube Movie Trailer Data**: YouTube data (likes, dislikes, comments, etc.) for movie trailers can be used in movie recommendation systems (Roy and Ding, 2020). We expand on this by using the trailer video itself, though metadata can be used in future work to provide better recommendations.

5.2 Limitations

Our approach has several notable constraints:

- **Dataset Size**: With 250 movies from only Paramount and 20th Century Studios, our dataset lacks the scale and diversity of commercial streaming catalogs, limiting genre coverage and under representing independent and international cinema.
- **Trailer Bias**: Trailers are marketing materials designed to maximize appeal, which may misrepresent actual movie content a horror trailer might emphasize jump scares while the film focuses on psychological tension.
- **Cold Start Problem**: Users must begin with a known movie they like; the system cannot make recommendations without a reference point or help users discover entirely new genres.
- **No Feedback Loop**: The system lacks mechanisms to learn from user interactions users cannot indicate which recommendations were successful or refine results based on their selections.
- **Audio Limitations**: Current audio embeddings capture low level acoustic features but miss higher level characteristics like musical genre, composer style, or dialogue quality that users may care about.

6 Conclusion & Future Work (0.5 page)

6.1 Key Findings and Contributions

This project successfully addressed the fundamental problem of user control in movie recommendation systems. Through IMDB 3.0 (Trailer IR), we demonstrated that multi modal trailer embeddings combined with user controllable weights provide a transparent and interactive alternative to traditional "black box" recommendations. **Research Question Outcomes:**

We achieved our three primary objectives:

1. **User Control**: Users can now explicitly manipulate recommendation criteria through slider controls across visual, audio, and narrative dimensions, providing direct influence over the recommendation process.

2. **Interpretable Embeddings**: Our CLIP, Librosa, and Gemini+BERT embeddings capture genuinely distinct and interpretable aspects of movies, allowing each modality to represent different facets of film content.

3. **Interactive Discovery**: The web platform at `https://coolmovievectors.com/` enables exploratory discovery that addresses the "doom scrolling" problem on streaming services by giving users agency in the search process.

Key Insights:

Our analysis revealed two important findings about trailer based recommendation. First, visual embeddings capture artistic style remarkably well cinematography, color grading, and visual aesthetics cluster meaningfully in the CLIP embedding space, allowing users to find movies with similar "looks" regardless of genre. Second, trailers encode substantial narrative information; our Gemini generated descriptions successfully captured plot themes, emotional arcs, and storytelling approaches, demonstrating that trailers reveal more story structure than initially expected. This validates using trailers as a rich source of multimodal movie features rather than requiring access to full films.

Technical Contributions:

We created a custom dataset of 250 official movie trailers with complete metadata, implemented user controllable late fusion for interpretable similarity search, and built an accessible web interface that makes vector search intuitive for non technical users.

6.2 Future Work

Several promising directions could extend this work:

Review Based Tagging System: The most immediate enhancement would be adding user generated tags derived from movie reviews. By mining common descriptive phrases from reviews (e.g., "mind bending," "slow burn," "visually stunning," "character driven," "plot twist"), we could create a fourth searchable dimension. Users could then filter by specific attributes: "Find me a sci fi movie that reviewers describe as 'philosophical' with similar visuals to Arrival." This would bridge the gap between our content based embeddings and the subjective qualities that viewers actually care about.

Dataset and Feature Expansion:

- Scale to 1,000+ movies covering diverse studios, genres, and decades
- Enhance audio analysis with music genre classification and dialogue sentiment
- Include cast, director, and production metadata

Personalization:

- Implement user profiles that learn preferred weight combinations over time
- Add feedback mechanisms (like/dislike) to refine recommendations
- Enable natural language queries: "movies like Blade Runner but with lighter tone"

Evaluation: Conduct formal user studies comparing our system to traditional recommendation interfaces, measuring satisfaction, discovery diversity, and time to decision metrics.

6.3 Closing Remarks

Movie recommendation is fundamentally about helping people discover content that resonates with aspects they may not be able to articulate upfront. By decomposing movies into interpretable dimensions and giving users control, we've created a more transparent and exploratory recommendation experience. IMDB 3.0 demonstrates that trailers are information rich sources for multimodal analysis, and that user controlled vector search can address the limitations of passive algorithmic feeds. We hope this work inspires a new paradigm where users are active participants in content discovery rather than passive recipients of opaque suggestions.

Acknowledgments

References

Chengbo Zheng, Yuanhao Zhang, Zeyu Huang, Chuhan Shi, Minrui Xu, and Xiaojuan Ma. DiscipLink: Unfolding Interdisciplinary Information Seeking Process via Human-AI Co-Exploration, August 2024. URL `http://arxiv.org/abs/2408.00447`. arXiv:2408.00447 [cs].

A Appendix: Additional Visualizations (Optional)

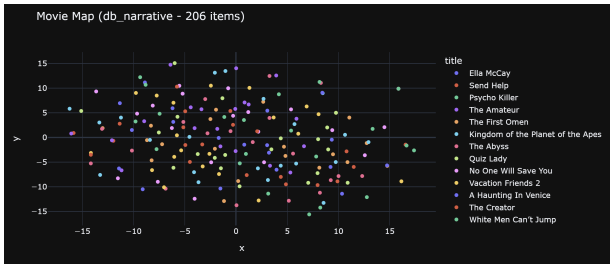


Figure 1: A 2D mapping of all the visual movie vectors

B Appendix: Code Implementation Details (Optional)